

RESEARCH ARTICLE

Research Misconduct in the Global South

Enhancing Academic Integrity: Drivers, Barriers, and Governance Pathways in China

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ABSTRACT

In recent years, the rapid growth in global research output has been accompanied by an increasing challenge of research integrity. This study, taking China as an example, systematically examines the drivers, barriers, and governance strategies of research integrity through policy text analysis and in-depth interviews, revealing a complex network of problems composed of institutional pressure, alienation of evaluation systems, and technical challenges. It finds the culture of 'publish or perish' resulting from a quantitative assessment system, uneven distribution of research resources, and lack of enough academic ethics education to be the main barriers to research integrity. Although the government has established a multi-level governance system, policy implementation still faces challenges. With the rapid development of AI technology, new forms of academic misconduct pose challenges to the traditional regulatory model. Based on the research findings, this paper proposes suggestions to promote research integrity, ranging from passive compliance to active self-discipline.

1 | Introduction

Academic misconduct has become a major challenge for the global scientific community. It is defined by the international academic community as the existence of fabrication, forgery or plagiarism in the application, implementation, review and report of research results, the presence of which violates the norms of academic integrity (Holtfreter et al. 2019). Academic misconduct involves two concepts which are academic integrity and research integrity. The former is defined as applying to student behaviour such as undergraduate thesis plagiarism, and the latter to professional researchers' conduct such as data falsification in projects. The core of this study is research integrity.

Recently, the complexity of the academic publishing system and globalisation of research activities has led to academic misconduct becoming more concealed and technologically advanced. Its manifestations have expanded from traditional data falsification, plagiarism, and duplication to manipulation of the peer

review process and systematic manipulation of data, tampering with and forging images (Kearney et al. 2024), and new academic ethical dilemmas such as using generative AI. An example is the experimental artificial trachea transplantation study published in *The Lancet* by Paolo Macchiarini; he was later found to have omitted pertinent data, resulting in the deaths of six patients (Else 2021). From 2011 to 2020, the share of papers retracted globally due to academic misconduct increased from 14.4% to 53.2% (Retraction Watch Database 2023). In 2023 alone, more than 10,000 research papers were retracted globally (Van Noorden 2023).

Academic misconduct in Asian countries shows significant regional commonality, with many cases being associated with career advancement pressure and flaws in research evaluation systems. The Japanese academic community has been criticised for systematic fraud and research pressure in which a typical case involved the fabrication of data for 11 figures in a paper published in *Stem Cell Reports* (Kyoto University 2018). In India,

plagiarism is primarily driven by the lack of adequate funds and pressure to publish articles (Kar 2024). According to the National Research Foundation of Korea's annual statistics, 544 cases of research integrity violations were recorded between 2015 and 2019. The three most common categories were invalid authorship, in which attributing authorship to people who had not contributed substantially or had no responsibility for the work (210 cases, 38.60%), plagiarism (174 cases, 31.99%), and redundant or duplicate publication (71 cases, 13.05%; Chou et al. 2024).

China's output of scientific knowledge has grown enormously since the turn of the century and it is now ranked first globally in terms of publication. Owens (2024) notes that in 2022, China overtook the United States for the first time in the Nature Index for contributions to natural sciences and that Digital Science's Dimensions database listed almost 830,000 papers. However, the accompanying issue of academic misconduct has increased and institutional retractions have become more prominent. As of December 2023, 18,742 retractions from journals indexed in the Web of Science database had been recorded, with China being the second most frequent country of origin (4621% or 24.6%) after the United States (5492% or 29.3%) (Retraction Watch Database 2023). While the government has established multi-level governance mechanisms, existing practices face two critical gaps that are understudied in English research. The first is that the drivers of research misconduct interact to form a circle, and they manifest in both institutional policies and researchers' daily practices. The second is emerging challenges from AI technology that outpace the coverage of traditional regulatory frameworks. This study addresses these gaps by suggesting integration of macro policy and micro practice, integration of traditional challenges and emerging risks such as AI-assisted misconduct, and integration of cultural elements such as Confucian integrity norms with modern governance tools.

2 | Literature Review

2.1 | The Influence of Research From China in Global Innovation

With one of the fastest increases in innovation capacity in the past 10 years, China has 26 of the world's top 100 scientific and technological innovation clusters, up from 24 in 2023 (World Intellectual Property Organization 2024a). The latest data shows that China's total social research and development expenditure in 2024 was US\$ 496.7 billion, up 8.3% from the previous year, the second largest in the world (National Bureau of Statistics 2025). A total of 272,600 international patent applications were submitted through the Patent Cooperation Treaty in 2023, with China submitting the most: this was the fifth consecutive year that China had topped the list since it first did so in 2019 (World Intellectual Property Organization 2024b).

China contributed 31% of the world's top 1% most-cited scientific output from 2018 to 2020 (Pisani et al. 2024). In 2024, the Nature Index ranked China first overall in terms of papers contributed to high-quality journals, leading in chemistry, earth and environmental sciences, and jumping to second place in life sciences (Baker 2024). In addition, Chinese scholars' share of highly cited papers is close to that of research powerhouses such

as the United States. For example, in the top 10% of highly cited papers in clinical medicine, although China's share is slightly lower in total, its efficiency relative to its total research output is on a par with international levels (Ganpatsingh 2025). China's research impact has been steadily increasing since 2000 and is now equal to much of Europe and the US. China's contribution to high-impact papers based on citation rate is also comparable to that of other research-intensive countries, even in disciplines where there is less research from China (Springer Nature 2024). However, this leapfrog development in scientific research output has exacerbated the phenomenon of quantity over quality. On April 20, 2017, *Tumour Biology* carried out a one-time retraction of 107 papers from China mainly because of false peer reviews (Yang et al. 2024). At the academic level, academic misconduct undermines the rigour of scientific research. This not only leads to the waste of scientific research funds but also affects the output and dissemination of scientific research achievements.

Chinese traditional culture has strict requirements for the conduct of scholars and regards academic integrity as the foundation of their existence. Confucianism has long dominated the academic ethics system in Chinese society, and its core of integrity is regarded as the cornerstone of personal cultivation and social relations. Academic misconduct affects the formation of academic values, and in the erstwhile Imperial examination system, if any signs of cheating were discovered, both the student and the staff member were punished (Sendur 2024). For instance, the Confucian classics emphasise: "If a person lacks trustworthiness, I don't know what s/he can be good for", elevating trustworthiness to the level of "the foundation of one's character" (*The Analects of Confucius: Governing*), and regard moral self-discipline as the basic code of conduct for intellectuals.

3 | Methodology

This study employs qualitative research to understand the causes of scientific research misconduct in China. This approach enables researchers to explore any interconnected meanings that would remain inaccessible through numerical analysis by elucidating micro-level dynamics that demonstrate how specific phenomena function within macro-level frameworks (Shaw 2016). Data was collected via document analysis and interviews. Document analysis was used to examine national policies and universities' responses, the documents being sourced from authoritative platforms such as the China Government Network, Ministry of Science and Technology Database, and Ministry of Education Policy Library, whose contents align with the study's focus on national-level governance frameworks. The objective was to establish the government's governance framework and stance on addressing research misconduct. The selective choice of policy documents followed a rigorous process based on significance and thematic relevance. Documents were selected if they were issued by top-level decision-making and governance bodies. Additionally, documents explicitly mentioning keywords such as 'research integrity' or 'academic misconduct' in their titles were prioritised.

Interview participant selection was strategically aligned with the research aims to harmonise specialised expertise with diverse viewpoints, ensuring both depth and breadth

in the exploration of the subject (Adeoye-Olatunde and Olenik 2021). The study's purposive sampling methodology recruited 36 participants possessing diverse backgrounds that aligned with the research objectives, with 29 participants holding doctorates from top 100 global universities. The disciplinary breakdown within STEM included 9 from life sciences, 8 from engineering, 7 from computer science, 5 from chemistry, 4 from mathematics, and 3 from environmental science, ensuring coverage of high-risk fields. Career stages were varied, comprising 11 senior professors, 15 associate professors, and 10 assistant professors in order to capture different institutional power dynamics. All participants understood the research purpose, potential risks, and the right to withdraw at any time. This study was approved by the Research Ethics Committee of the university of the first author.

Recruitment proceeded through snowball sampling and semi-structured interviews served as the methodological framework, enabling systematic investigation of participants' policy implementation practices and institutional constraints. This design balanced structured inquiry with adaptive probing (Magaldi and Berler 2020), thereby revealing the dynamics of scientific research integrity management.

We utilised thematic analysis as our core methodological approach for examining qualitative data. Thematic analysis facilitates the systematic identification and interpretation of recurring patterns within datasets, enabling researchers to synthesise shared participant experiences while preserving methodological precision (Braun and Clarke 2012). Following Braun and Clarke's (2006) six-phase framework, the analytical process comprised data immersion, initial code generation, theme development, iterative theme refinement, final theme conceptualization, and comprehensive reporting. This rigorous

structure ensures both methodological transparency and the capture of participants' multifaceted perspectives through organised data exploration.

Table 1 is the example of the thematic analysis. We applied Braun & Clarke's six-phase framework with NVivo 12: two researchers independently coded the interview transcripts, generating 187 initial codes that were refined into the 4 core themes presented in the 'Findings'; a Cohen's kappa coefficient of 0.85 confirmed excellent inter-coder consistency, with a third researcher resolving discrepancies. Interviews were conducted in Mandarin, the participants' native language to ensure authenticity, and coding was first done in Mandarin to preserve contextual nuances; after this, an English translation adopted a 'double-verification' process—bilingual researcher translation paired with back-translation, cross-checked against original interview audio, to maintain accuracy, critical for capturing the China-specific insights outlined in the original transcripts.

4 | Findings

4.1 | Tensions in Quantitative Evaluation Systems and Institutional Pressure

The 'publish or perish' quantitative evaluation system is one of the major causes of academic misconduct. Chinese universities often use the number of papers listed by the Science Citation Index as a core indicator for professional title evaluation. To obtain tenure, applicants need to publish a certain number of research papers within a certain number of years in specified internationally indexed journals (Tian et al. 2016), and physicians in China also have this pressure because they need to publish research articles to gain promotion (Else and

TABLE 1 | Thematic analysis.

Core theme	Sub-themes	Example initial codes
1. Tention in quantitative evaluation system and institutional pressure	– SCI-centric promotion criteria – Performance bonuses tied to publication quantity – Publication-oriented and paper-centric chain	– “SCI papers are the core indicator for professional title evaluation” – “Performance indicators are directly linked to bonuses” – “Third-party paper revision services”
2. Insufficient oversight and punishment	– Formalistic peer review – Uneven punishment (leniency for senior scholars) – Symbolic penalties	– “Peer review has become a formality” – “Top scholars are rarely punished for misconduct” – “Punishment is limited to restricted promotion”
3. Lack of academic ethics education	– Slogan-like ethical training – Lack of practical guidance – Institutional reputation prioritised over justice	– “Ethics education only emphasises ‘no plagiarism’” – “No enough training on data falsification boundaries” – “Universities cover up misconduct to protect reputation”
4. Resource concentration in a pragmatically oriented academic ecosystem	– Title-based resource allocation – Short-term output pressure – Speculative resource competition	– “Titles determine access to project funds” – “To pursue quick results” – “Reusing research ideas for multiple fund applications”

Van Noorden 2021). A survey of 163 dishonesty research papers reported in the medical research integrity bulletin of the National Health Commission found that paper trading and writing and submission services accounted for as much as 50.3% of these dishonesty research papers (Cai and Xu 2022). In other words, this 'publish or perish' system has led to the growth of the 'paper factory' to some extent, and the growth in transaction volume of writing and submission services on e-commerce platforms.

We need to deal with the annual performance indicators as publishing papers is a performance indicator, some of which were directly linked to performance bonuses.

(Interviewee 1)

A quantitative assessment system distorts the long-term public value of knowledge innovation into short-term performance indicators, forcing scholars to view paper production as a survival tool and giving rise to the distorted behaviour of research for publication. When the number of papers and their impact factor become important measure of a scholar's value, academic achievements become 'piece-rate commodities', researchers would shorten cycles, chase hotspots, and even resort to data falsification in pursuit of short-term gains. This evaluation system simplifies multiple academic values to a single rule of 'publication means survival,' warping scholars from knowledge creators to production machines to some extent, with their identities different from the original intention of academic research.

Academic misconduct should not be attributed to personal misconduct. If the research assessment mechanism emphasizes quantitative indicators, it may push some researchers to be out of balance between academic ethics and assessment pressure.

(Interviewee 12)

Research misconduct in the scientific research ecosystem is one of the results of the tension between institutional requirements and the logic of individual survival. It has developed a inertia as, when promotions and project applications are bound to quantitative indicators, the evaluation of the academic community might become a numerical game, and individuals might pay more attention on quantity over quality in the face of institutional pressure.

4.2 | Insufficient Oversight and Punishment

In the current academic governance system, the review mechanism for academic misconduct is sometimes not fully rigorous, and punishments are often mitigated for the sake of the institutions' reputation. The disciplinary system, which is supposed to be centered on maintaining academic integrity, tends to become symbolically flexible management in practice, resulting in a potential lowering of the cost of violations.

An academican who was exposed for using unproven pictures in a paper was criticised on the grounds of 'misuse', but if it were a young researcher, the treatment would be more strict.

(Interviewee 3)

It is not common to punish top scholars for academic misconduct but for ordinary scholars.

(Interviewee 4)

Insufficient punishment from academic institutions may result in subject communities within academia forming relatively protective connections due to interest-related factors, leading peer review and data verification to become not fully rigorous. The relative leniency of the review process and inadequate procedural justice mean academic misconduct may remain unaddressed despite procedural compliance. It is also worth noting that insufficient punishment makes the cost of academic misconduct relatively lower than the potential benefits of violation.

When academic misconduct comes only at the cost of restricted promotion, but are not punished as they should be, it reduces the cost of mistakes and is not conducive to the healthy development of the academic ecosystem.

(Interviewee 15)

There is a serious institutional divide between the current academic supervision mechanism and substantive accountability. This disconnection is reflected not only in the lag and ambiguity of the disciplinary procedures, but also in a 'high and low' logic of punishment. On the one hand, a zero-tolerance stance is taken toward academic misconduct; On the other hand, by reducing the severity of penalties, extending the investigation period, and blurring the boundaries of responsibility, some punishment is reduced to a symbolic governance performance. This accountability mechanism leads to an reverse incentive structure. Because the risk cost of academic misconduct is depressed while compliant research requires huge time and intellectual cost, speculators might face with a situation where the benefits of violation are higher than those of compliance.

4.3 | Lack of Academic Ethics Education

In the current academic ecosystem, some researchers lack systematic training in academic norms and integrity awareness, and this has become a factor restricting the improvement of academic quality.

Ethical education in some universities remains at the surface, with students need being taught with systematical academic norms.

(Interviewee 18)

When we were students, my supervisor repeatedly emphasize the academic bottom line of no fraud and warn of the lifelong negative effects of academic fraud, but some research environments may not attach enough importance to such norms.

(Interviewee 9)

Academic integrity education in universities and research institutions is sometimes a formality, lacking a targeted training system and failing to cover the key stages of researchers' career development. Some existing training neglect practical guidance on specific operational boundaries such as data falsification and citation norms.

Some universities in an effort to maintain universities' reputation, take self-protective measures regarding the involved individuals.

(Interviewee 7)

When higher education institutions, which are supposed to be guardians of academic integrity, prioritise protecting own reputation above academic justice, this creates institutional loopholes in the supervision of academic misconduct. When plagiarism, fraud and other acts are regarded merely as 'public opinion risks' rather than academic red lines, the disciplinary mechanism becomes a formality.

The frequent academic misconduct is related to the fast pace of research and loose team management.

(Interviewee 22)

The utilitarian orientation of academic achievements and short-term incentive mechanisms have further weakened researchers' sense of academic ethics. Under the pressure of competition, some scholars, in pursuit of quick output, have turned academic misconduct into a strategic means of survival. Meanwhile, the neglect of integrity education has marginalised academic norms to secondary professional requirements rather than core values. This weakening of values blurs the ethical perception of researchers, promotes the growth of strategic breach of trust behaviours.

Academic ethics are the fundamental code of conduct for researchers, and it is crucial that to teach academic ethics for our future researchers.

(Interviewee 9)

Therefore, the lack of formalisation of academic ethics education constitutes one of the causes of the academic integrity crisis. When researchers lack academic ethics education and training and are pressured to produce quick and easy research, technical fraud is sometimes accepted as a choice. Only by transforming academic ethics into actionable guidelines can norms be internalised as researchers' conscious choices and the ethical foundation for high-quality academic development be firmly established.

4.4 | Resource Concentration in a Pragmatically Oriented Academic Ecosystem

In the current pragmatically oriented academic ecosystem, research management systems are increasingly intensifying resources through outcome indicators. Core resources such as project funds and experimental equipment are directly linked to short-term indicators such as the number of papers and impact factors. This pragmatically oriented academic ecosystem of determining resources by results traps grassroots researchers in a circle where it is difficult to develop research without resources and even difficult to obtain resources without results.

The research evaluation system is utilitarian, such as taking the publication of high-index papers as the core indicator and dividing the author's identity into multiple grades, such as 'first author,' 'co-first author,' 'corresponding author,' 'co-corresponding author,' etc.

(Interviewee 21)

The paper-oriented assessment mechanism pushes scholars to pursue quick success, and under the pressure of promotion, academic misconduct is possible.

(Interviewee 13)

Academic achievements are tied to social status and economic gains, and instrumental rationality overpowers value rationality. This pursuit of utilitarianism in the academic ecosystem has enabled academic misconduct to evolve from moral misconduct to the strategic choice of rational economic man. When high returns are expected, such as tenure as a result of top journal papers, some scholars might take risks and distort academic integrity into tradable credit derivatives through data tampering and other means.

The evaluation system is overly dependent on academic titles, which squeezes the development space of ordinary researchers, also reinforces the hierarchical differences in salary, academic resources and social recognition.

(Interviewee 12)

Some scholars apply for funds by using the same research ideas which were packaged into different projects and funds were applied for through multiple channels.

(Interviewee 18)

The 'title first' promotion rule, which is widespread in universities, requires production of landmark achievements such as top journal papers, major projects, etc. This pushes scholars to engage in speculative behaviour such as rapidly accumulating

achievements to compete for limited resources. The four themes identified above do not operate in isolation but form a self-reinforcing cycle perpetuating research misconduct in the academic ecosystem. At the cycle's core, the quantitative evaluation system, centered on 'publish or perish' and SCI paper counts, directly fuels research misconduct. As Interviewee 21 notes, universities tie "high-level paper publication" to author grade divisions and resource allocation. Researchers, in particular some junior researchers, lack the resources to conduct rigorous long-term research, and have incentives to engage in misconduct to meet quantitative targets.

Compounding this cycle, insufficient institutional oversight and punishment reduce the cost of such misconduct. The leniency toward senior scholars and symbolic penalties, like Interviewee 15's mention of "restricted promotion but no financial sanctions," fail to deter violations and embolden researchers to prioritise short-term gains over ethics. This dynamic is amplified by the lack of academic ethics education. As Interviewee 18 notes, ethics training is often reduced to no plagiarism. This leaves researchers, especially early-career scholars, without clear guidance to navigate the tension between evaluation pressure and integrity. Without internalised ethical norms, researchers may view misconduct as a "strategic survival tool," a phrase used by Interviewee 22, amid limited resource and quantitative demands. Overall, insufficient oversight and ethics education remove barriers to such behaviour, and the resulting misconduct further entrenches the 'quantity over quality' logic that sustains the evaluation system and resource concentration. This process creates a feedback loop that undermines long-term research integrity.

4.5 | Current Policies on Research Misconduct

China's academic misconduct governance system is characterised by significant historical continuity, full-cycle coverage, and institutional systematicness. Since the introduction of the first *Measures for Handling Research Misconduct in the Implementation of the National Science and Technology Program* in 2006, China has built a regulatory framework covering the entire chain of research activities through multi-level policies (Ministry of Science and Technology 2006). It covers key areas such as project applications, publication of research results, and degree awarding. This system focuses on addressing current issues and reflects long-term institutional determination. The continuous strengthening of standards through policy iterations from 2011 to 2025 (such as the 2024 *Guidelines on the Use Boundaries of AIGC [Artificial Intelligence-Generated Content] in Academic Publishing 2.0*) demonstrates that China's academic integrity building is not a short-term movement (Institute of Scientific and Technical Information of China 2024), but rather embedded in a long-term governance mechanism. More importantly, governance extends to the early stages of academic growth. Through documents such as the *Measures for Random Inspection of Doctoral and Master's Theses* (2014) and *Measures for Random Inspection of Undergraduate Graduation* (2020), academic norms have been extended to postgraduate training and undergraduate education, forming a dual-track system of degree acquisition and career development constraints, fundamentally curbing the risk of intergenerational transmission of

academic misconduct (Ministry of Education 2014, 2020). In addition, the *Multi-Departmental Collaborative Rules on the Investigation and Handling of Research Misconduct* (Ministry of Science and Technology 2022) cover deep-rooted issues such as funding abuse and ethical review loopholes, forming a joint supervision network of 22 institutions including the Ministry of Science and Technology and the Ministry of Education. In response to the risks of emerging technologies, the Chinese Academy of Sciences issued an integrity reminder on the standardised use of artificial intelligence technology in research activities, explicitly prohibiting the use of AI to generate false data or papers (Research Ethics Committee of the Chinese Academy of Sciences 2024). Table 2 lists the measures that Chinese government agencies have taken since 2006 to fight scientific fraud and academic misconduct and push for better science management. In terms of validity level in China, it can be classified into Laws, Departmental Regulations, Normative Documents, and Industry Standards.

The *Degree Law of the People's Republic of China*, passed in April 2024, for the first time included academic misconduct such as ghostwriting, plagiarism, and data fabrication within the scope of compulsory punishment at the legal level, clearly stipulating that universities have the right to revoke the qualifications of those involved (The Standing Committee of the National People's Congress 2024). In the same year, the revised *Regulations of the National Natural Science Foundation of China* strengthened the research ethics system and the joint disciplinary mechanism, imposed penalties such as funds recovery and project qualification cancellation on offenders, and published details of misconduct cases (State Council 2024). The National Natural Science Foundation of China, for example, reported over 60 misconduct cases in the period 2023–2025 (Table 3), covering the entire chain of dishonest behaviours such as data forgery, image tampering, proxy writing and submission, and review fraud. The handling measures strictly follow the four-step disciplinary system of revocation of project, recovery of funds, disqualification and public notification.

4.6 | Institutional Responses to Research Integrity

Chinese universities and research institutions have adopted proactive measures to align with national policies. The C9 League, a consortium of China's top nine universities, has taken the lead in implementing specialised management measures for research integrity. Nationally, policy implementation varies significantly due to institutional resource disparities. C9 universities can implement national policies at high standards because they have sufficient funding for technical tools and full-time staff. For example, Tsinghua University has established a comprehensive mechanism for investigating academic misconduct throughout the entire research process to prevent and address such behaviours, and Fudan University introduced China's first institutional ban on AI-generated content in academic settings, setting a regulatory precedent for technological ethics standards. In contrast, universities with little influence cannot afford technical supervision systems or bear the implicit costs of punishing senior scholars (e.g., losing discipline leaders and/or projects, or declining rankings), leading to selective implementation and relaxed penalty standards.

TABLE 2 | Laws, regulations and policy documents on academic misconduct.

Document	Validity level	Governance bodies
Degree Law of the People's Republic of China	Laws	The Standing Committee of the National People's Congress (2024)
Guidelines on the Use Boundaries of AIGC in Academic Publishing 2.0	Industry Standards	Institute of Scientific and Technical Information of China (2024)
Integrity Reminder on Regulating the Use of Artificial Intelligence Technology in Scientific Research Activities	Industry Standards	Research Ethics Committee of the Chinese Academy of Sciences (2024)
Detailed Rules for the Investigation and Handling of Academic Misconduct in Institutions of Higher Learning	Departmental Regulations	Ministry of Education (2024)
Code of Conduct Guidelines for Responsible Research (2023)	Industry Standards	Ministry of Science and Technology (2023)
The Multi-Departmental Collaborative Rules on the Investigation and Handling of Research Misconduct	Departmental Regulations	Ministry of Science and Technology (2022)
Measures for Handling Research Misconduct in National Natural Science Foundation Projects	Departmental Regulations	National Natural Science Foundation of China (2023c)
Several Opinions on Further Strictly Regulating the Quality Management of Degree and Graduate Education	Normative Documents	Degree Committee of The State Council and Ministry of Education (2020)
Measures for Random Inspection of Undergraduate Graduation	Departmental Regulations	Ministry of Education (2020)
Several Opinions on Further Strengthening Research Integrity Building	Normative Documents	CPC Central Committee and State Council (2018)
Measures for the Prevention and Handling of Academic Misconduct in Institutions of Higher Learning	Departmental Regulations	Ministry of Education (2016)
Guiding Opinions of The General Office of the State Council on Optimising the Academic Environment	Normative Documents	General Office of the State Council (2015)
Measures for Random Inspection of Doctoral and Master's Theses	Departmental Regulations	Ministry of Education (2014)
Measures for Dealing with Academic Thesis Fraud	Departmental Regulations	Ministry of Education (2012)
Opinions of the Academic Degrees Committee of The State Council on Strengthening the Construction of Academic Ethics and Academic Norms in the Work of Degree Conferral	Normative Documents	Academic Degrees Committee of The State Council and Ministry of Education (2010)
Notice of the Ministry of Education on Serious Handling of Academic Misconduct in Institutions of Higher Learning	Normative Documents	Ministry of Education (2009)
Measures for Handling Research Misconduct in the Implementation of the National Science and Technology Programs	Departmental Regulations	Ministry of Science and Technology (2006)

In recent years, the frequent high-profile cases of academic misconduct have highlighted the urgency for effective governance of scientific research integrity. In 2019, one popular actor raised public doubts about his academic background as a result of a comment regarding the Chinese academic database CNKI during a live stream, and investigations revealed that one article was published during his doctoral studies but almost 40% of which was plagiarised (Zhang 2024). Not only did the Beijing Film Academy revoke his doctoral degree after verifying this, Peking University's Guanghua School of Management also cancelled his postdoctoral status (Ministry of Education 2019). The

2017 retraction of more than 100 Chinese scholars' papers at one time (Koo and Lin 2024) led to an investigation which found that these papers were suspected of falsifying peer reviews, data fabrication and being written and submitted by third parties. After the incident was exposed, the Ministry of Science and Technology (2017), in conjunction with multiple departments, launched a joint investigation, suspended the national science and technology projects being undertaken by the individuals involved, revoked their academic awards and honorary titles, and added the relevant responsible entities to the blacklist of scientific research integrity.

TABLE 3 | Misconduct cases identified by the National Natural Science Foundation (compiled by the author based on the official website of the National Natural Science Foundation).

Batch of publications	Number of cases
First Batch of Research Misconduct Case Handling Results Announcement (2025)	15
Second Batch of Research Misconduct Case Handling Results Announcement (2024a)	10
First Batch of Research Misconduct Case Handling Results Announcement (2024b)	12
Second Batch of Research Misconduct Case Handling Results Announcement (2023a)	24
First Batch of Research Misconduct Case Handling Results Announcement (2023b)	6

Another typical case was that in 2020, when the team of one academician was revealed to have repeatedly used the same data in several papers. The Ministry of Science and Technology (2021a, 2021b) eventually evaluated that no subjective fraud was found, but pointed out that some papers exhibited improper academic behaviour. The incident sparked a widespread public debate about research norms for top scholars. Such high-profile cases, however, reflect a responsive governance logic driven by public outcry. The 2019 the actor case which triggered national doubts about doctoral training quality and the 2020 academician data reuse incident which aroused public concerns over top scholars' ethics resulted in public penalties such as degree revocation and project cancellation primarily to safeguard academic credibility amid public pressure.

The government and universities have thus demonstrated multi-level governance in the fight against academic misconduct. The Ministry of Science and Technology (2021a, 2021b) has adopted a zero tolerance attitude toward such high-impact cases through the co-work mechanism for research integrity building, and the public announcement of investigation results through the China Research Integrity Network to create a deterrent effect. For example, in 2024, the National Natural Science Foundation of China (2024a, 2024b) published the names of 21 scholars involved in behaviours such as trading data and duplicate applications, cancelled their projects, took back their funds, and disqualified them from applying for further funding for three to 5 years.

In 2018, the *Several Opinions on Further Strengthening Research Integrity Construction* (CPC Central Committee and State Council 2018) proposed a 'lifetime accountability' mechanism which requires that academic misconduct be directly linked to professional title promotion and project applications. The introduction of the *The Research Integrity Investigation Processing Rules* (Central People's Government 2019) has clarified the standardised process to handling research misconduct. In addition,

a blacklist of lower quality journals was established which made it clear that publishing in these would not count toward research performance, in order to curb the phenomenon of 'paper flooding' at the source (Ministry of Education 2021).

In recent years, the government has also established an information sharing mechanism for academic misconduct. This requires universities to adopt a one-vote veto system for those involved and impose strict restrictions to professional title evaluations and project applications. Universities, as the entities directly responsible for academic misconduct, have achieved a closed-loop governance by establishing a full-chain system of 'prevention, supervision and punishment.' According to the documents *The Detailed Rules for the Investigation and Handling of Academic Misconduct in Higher Education Institutions* (Ministry of Education 2024), universities are required to establish an academic committee as the highest investigation and evaluation body, set up a specialised executive body to handle reports, form investigation teams and promote an ethical academic atmosphere. This document requires universities to disclose the results of their academic misconduct investigations within a certain timeframe for public information and to restrict the participation of the guilty parties in research project applications and bar them from the awarding of honorary titles.

China's crackdown on academic misconduct has been internationally recognised, leading to a 2018 *Nature* article which acknowledged that "China sets a strong example on how to address scientific fraud." The government and universities are building a scientific ecosystem in which researchers do not dare to break trust and are not able to break trust through institutional improvement and enforcement rigidity. In the future, industry standards need to be further refined to promote the deep integration of academic ethics education and technology monitoring.

5 | Discussion

5.1 | Reform of the Evaluation System and Quality-Oriented Transformation

The current quantitative evaluation system centered on the number of papers published has shown obvious institutional flaws. The frequent occurrence of phenomena such as paper factories and data fraud essentially reflects the deviation of quantity over quality in scientific research evaluation. For example, most of the authors involved in the 107 papers retracted by the journal *Tumour Biology* in 2017 are from the field of clinical medicine (Yang et al. 2024). The system design that links their career advancement to the publication of SCI papers pushes researchers to shift their focus from clinical practice to paper production.

This study's findings concur with Holtfreter et al.'s (2019) global research. Through surveys of natural and social science scholars, they identified evaluation pressure as a core trigger for academic misconduct—a pattern prevalent across national research ecosystems. This study further uncovers the uniqueness in the Chinese context. China's recent 'breaking the five-only' reform (abandoning the orientation of 'only papers, titles, degrees, awards, and title') aims to alleviate this quantitative assessment

pressure. However, interview data shows reform effectiveness varies markedly as a result of institutional resource disparities: without mature alternative evaluation tools, publication counts remain a strict promotion requirement in most universities. Though some universities have adopted diversified evaluation, they still face the conflict between short-term performance and long-term innovation as Interviewee 13 noted, “evaluation inertia is hard to break.” Additionally, this study finds that resource monopoly in China’s research field is more focused on title-oriented allocation. Interviewee 12 stated that “over reliance on titles in the evaluation system constrains the development opportunities of ordinary researchers also exacerbates hierarchical disparities in research grants.” The unique structure of tight title and resource ties in China’s academic system underpins the contextual contributions of this research.

Reform requires breaking the path dependence of evaluating worth based purely on number of publications, and establishing a diversified evaluation framework that includes peer review, representative work system and social influence. On this basis, the shift to quality-oriented assessment requires the construction of a multi-dimensional academic value identification system which weakens the decisive role of the number of papers, and instead focuses on the originality of research, data transparency, and long-term academic contribution. In practice, a dual-track evaluation mechanism could be explored: for basic research, the evaluation cycle could be extended to 8 years or even longer, with a focus on examining the lasting influence of theoretical breakthroughs. For applied research, a joint review mechanism of laboratory and industry could be established, incorporating the effectiveness of technology transformation into the assessment to more accurately identify the true academic value of the research. The existing resource allocation mechanism is deeply entwined with quantitative assessment, resulting in the situation where the number of papers determines the amount of funds, which is particularly pertinent to young scholars. This study suggests that a quality priority channel be established in national + level scientific research projects, and stable funding for five to 10 years should be provided to selected research that might otherwise be refused as highly innovative and risky.

5.2 | Construction of Governance Regarding Academic Integrity

The rapid development of AI has brought about an unprecedented tool revolution in academic research, and AI is reshaping the boundaries of academic practice with efficiency and precision. However, this wave of technological empowerment has also brought new challenges to the academic field. The possibility of falsifying data, tampering with images, and even automating research results has made it exponentially more difficult to regulate academic integrity. Meanwhile, the instrumental rationality of AI technology further magnifies the drawbacks of the current academic evaluation system—quantitative assessment and short-term performance orientation force scholars to be receptive to the ethical compromise of technology doing the work for them.

AI is a double-edged sword in scientific research. It can both exacerbate academic misconduct due to the abuse of technology

and assist in detecting and curbing misconduct. The abuse of AI is giving rise to new patterns of academic misconduct—deep learning algorithms can generate realistic fake experimental data, and natural language processing technology can automate the production of the text content needed by ‘paper factories,’ greatly lowering the threshold for academic fraud. Meanwhile, AI provides revolutionary technological tools for academic governance. For example, literature screening systems based on machine learning can identify the mass production characteristics of paper factories, blockchain technology provides an immutable traceability mechanism for scientific research data, and big data analysis can build predictive models for academic misconduct. “It’s easy for AI to find out if the picture has been modified” (Interviewee 9).

The essence of this technological paradox lies in the application effect of AI in research, and ultimately depends on the value orientation of the institutional designer—only when ethical norms adapt in step with technological innovation, and when the technological governance system can effectively identify and curb the risk of AI abuse, can AI truly become an enabling tool for promoting research integrity rather than a technological accomplice that disrupts the academic ecosystem. The 2024 *Guidelines 2.0 on the Usage Boundaries of AIGC in Academic Publishing* established regulations to address the risks of AI technology abuse. Under the double tension of AI empowerment and regulation, academic integrity governance needs to reconfigure from individual ethical education to a systemic regulatory mechanism. “For our discipline, AI is acceptable as long as students seriously produce graphs and data, do practical work, and polish their language” (Interviewee 15).

Digital tools have lowered the technical threshold for data manipulation, but they have also provided new means for academic oversight. Similarity check rates—from Turnitin for student work in universities to the Crossref Similarity Check adopted by publishing groups such as Elsevier and Springer Nature—are generally used as the first threshold for manuscript screening. “Plagiarism checks are strict, including in the journal in which they are published. Now each journal has its own process of checking for plagiarism” (Interviewee 24). Some top journals even require authors to submit their original data for review. “Now there are individual researcher IDs, too, and it’s easy to find... That’s what your colleagues look at, so it’s clear to everyone what changes you’ve made” (Interviewee 18).

The popularity of unique identification systems such as ORCID and ResearcherID has enabled scholars to form a traceable knowledge graph of their academic output—any abnormal surge in output, change in research direction, or change in collaboration networks can be detected by peers through platforms such as Scopus and Web of Science. This ‘panoramic’ academic oversight environment is reshaping the paradigm for maintaining research integrity: although digital tools have lowered the technical threshold for data manipulation (such as Python scripts for batch-generating simulation data and GAN networks for synthesising false experimental images), countermeasures such as blockchain evidence storage and AI text analysis are evolving in parallel. Behind this technological game is essentially the transparency revolution that the academic community is seeking in its digital transformation. When every paper is exposed to open

source retrieval and the academic trajectory of each researcher can be traced by their peers, technology may eventually drive the paradigm shift of research integrity from passive compliance to active self-discipline.

5.3 | Suggested Improvements to the Academic Ethics Education System

Academic norm training needs to be strengthened further. The content of academic ethics courses shall pay more attention to the cultivation of deep-rooted values such as the responsibility of the academic community, data transparency, and cooperation norms. Ethical education needs to establish a stepwise training system and internalise academic norms into the conscious awareness of researchers. At the undergraduate level, practical training such as data recording and citation norms can be embedded through experiential courses. Postgraduate education can add discussions on academic ethics scenarios, such as case seminars on the boundaries of image processing; professional scientists, on the other hand, need to regularly participate in academic integrity and ethics workshops to learn about the cutting-edge standards of international academic integrity. In addition, drawing on the ancient Chinese tradition that ‘actions speak louder than words’ and the implementation of the ‘mentor responsibility system,’ requiring mentors to demonstrate honest behaviours such as data archiving and the public disclosure of negative results, would solidify academic integrity in the inheritance between teachers and students.

At the policy level, universities can be required to incorporate academic integrity and ethics courses into credit assessments and add an ‘academic integrity file’ to promotion evaluations to record the positive behaviours of scholars in participating in data sharing and correcting errors. From a technical perspective, with the assistance of artificial intelligence, a virtual digital platform could be developed to enable researchers to experience the effect of paper retraction on academic reputation. This closed-loop design of ‘education – practice – feedback’ can not only transform ethical norms from ‘external constraints’ to ‘internal recognition,’ but also promote the deep transformation of academic culture from ‘not daring to break trust’ to ‘not willing to break trust’ with the help of incentive mechanisms such as ‘integrity credits.’

6 | Conclusion

This study reveals a complex interplay between institutional pressures, cultural norms, and technological advances that collectively shape the landscape of research misconduct in China. At the institutional level, the pervasive ‘publish or perish’ culture, coupled with an over-reliance on quantitative metrics for career advancement, has created a system where academic success is often measured by output quantity rather than research quality. This pressure-cooker environment, as evidenced by our interviews with Chinese researchers, drives scholars toward meeting publication targets. The current evaluation system’s emphasis on short-term achievements is in conflict with the slow, iterative nature of genuine scientific discovery.

Academic misconduct has multiple causes. First, the evaluation system overly relies on the number of papers and impact factors. Added to this, the pressure of competition for academic resources, such as promotion, projects, and funding, further gives rise to short-sighted behaviours. Secondly, loopholes in the regulatory system are manifested in lax reviews and lenient punishment by academic institutions. In addition, researchers generally lack systematic training in academic norms and integrity awareness. More crucially, peer review and data verification become formalistic, and the problem of systematic fraud persists in journals and authoritative institutions. These factors together have shaken the foundation of academic integrity, causing damage to science’s credibility.

Since 2006, China’s policy responses to research misconduct have evolved significantly, transitioning from reactive measures to a more comprehensive governance framework. However, these policies face substantial implementation challenges. While top-tier universities have established rigorous review processes, resource disparities mean that enforcement remains inconsistent across institutions. Moving forward, China’s path toward research integrity requires multi-level reforms. Institutionally, evaluation systems shall shift from quantity-based to quality-focused metrics, incorporating measures of research impact and reproducibility. Culturally, academic institutions need to more systematically carry out ethical education, drawing on both traditional values and contemporary case studies. The international research community can contribute through collaborative efforts to establish global standards, recognising that research misconduct is a transnational challenge requiring shared solutions. Ultimately, building a research culture that truly values integrity over output will require sustained commitment from policymakers, institutions, and individual researchers alike.

Author Contributions

Xu Liu contributed to the conceptualization and methodology of the study and participated in writing review and editing. Jia Hao Luo performed formal analysis and visualization. Tan Yun was involved in data curation and writing review and editing. Xiang Lan handled data curation, visualization, and resources. This study is a phased achievement of the Shenzhen Municipal Science and Technology Strategic Research Project for 2025, titled ‘Research on Institutional and Mechanism Innovation for Promoting the Integrated Reform of Education, Science and Technology, and Talent’ (Project No. RKK20250526105006010).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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